Upstream and Downstream Impacts of College Merit-Based Financial Aid for Low-Income Students: Ser Pilo Paga in Colombia

HETEROGENEITY & ROBUSTNESS ANALYSIS

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Ser Pilo Paga

- English translation: being smart pays off
- Provides full scholarship loans for attending college for low-income, high-achieving students
- Developed in 2014 in Colombia
- Background:
  - Student loan and financial aid programs are underdeveloped in Colombia.
  - Reference: before SPP 11% of first-year undergraduate students had a loan. In 2017 70% US undergraduate students receive some form of financial aid

Ser Pilo Paga

- Eligible students:
  - Merit-eligibility: Score above a cutoff in the national standardized high school exit exam, SABER 11. (Being in the top 9%)
  - Need-eligibility: Score below a cutoff in the socioeconomic index SISBEN. (Belong to the poorest 50% of households)
  - Being admitted by a Higher education institution (HEI) with High Quality Accreditation.

Londoño-Vélez et al. (2020)

Part 1 - (my focus)

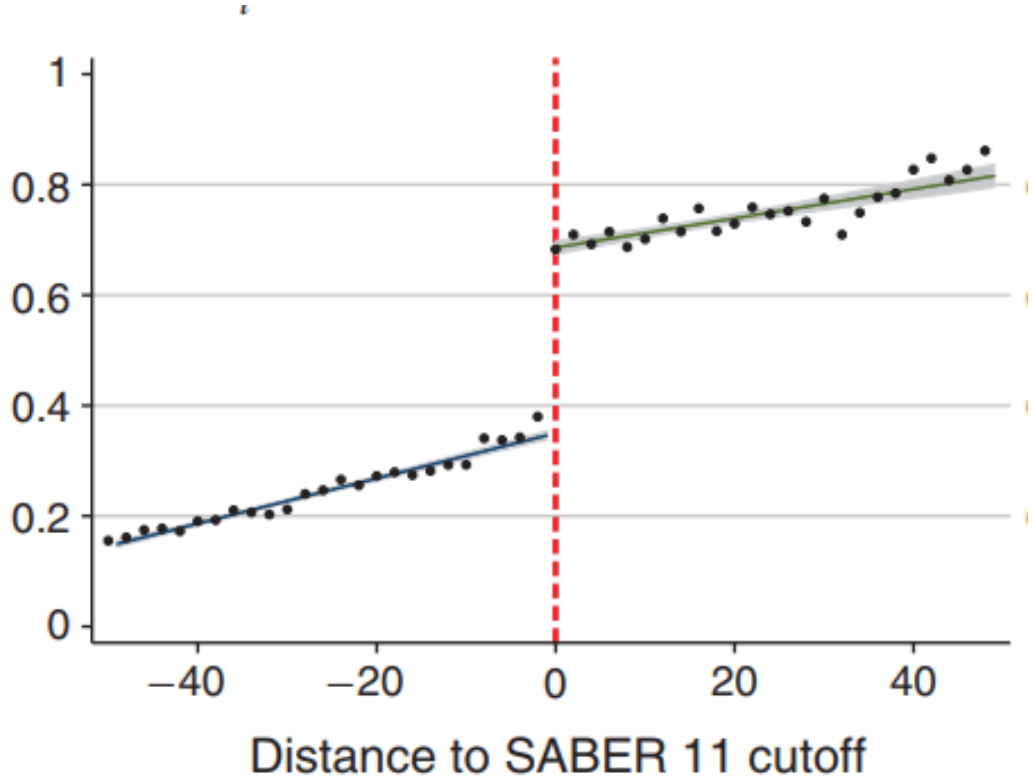
- Study the impact of removing financial constraints for low-income, high-achieving students in the immediate post-secondary enrollment
- Causal effect is identified through a Regression Discontinuity (RD) design.
- Administrative data from several sources

Part 2

- Study the broader effects on financial aid on the entire universe of students.
- Diff in Diff approach
- Not all data available

Londoño-Vélez et al. (2020)

Figure: Immediate Post-secondary Enrollment

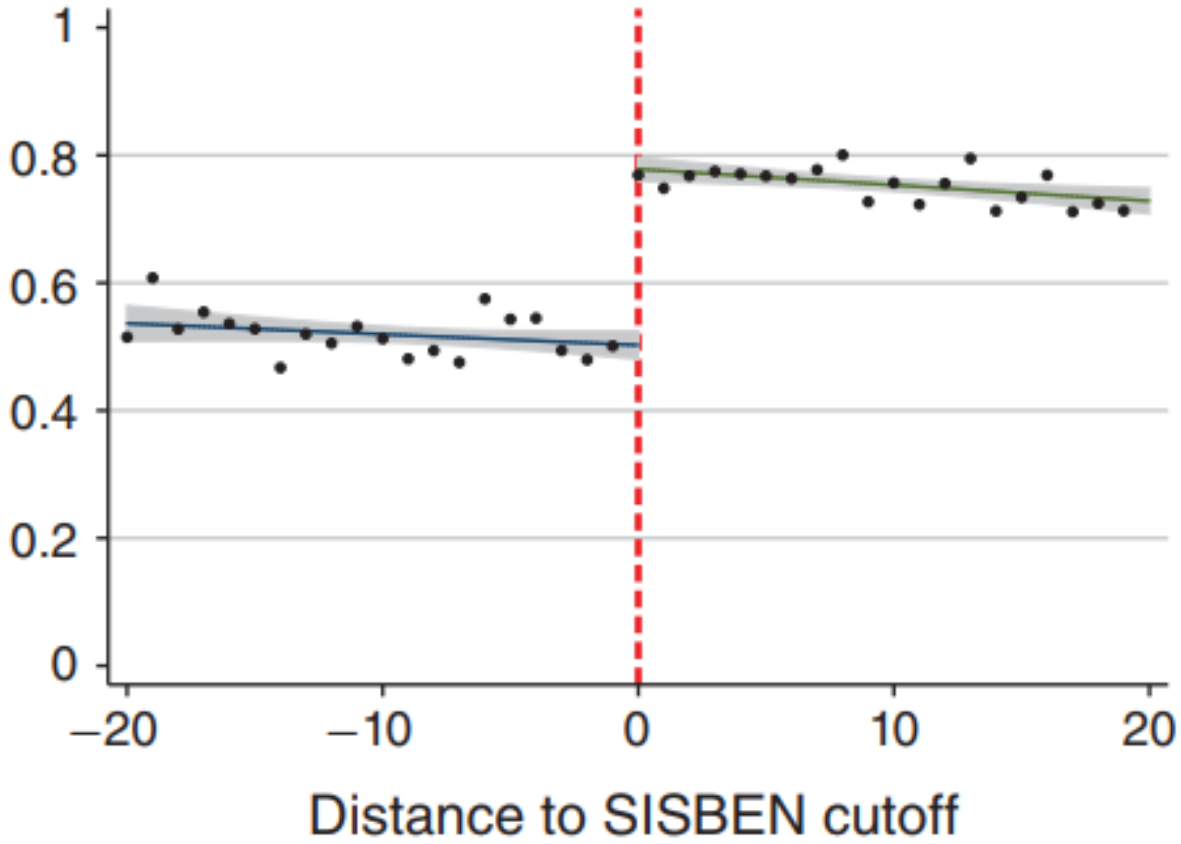


Source: Londoño-Vélez, Rodríguez, and Sánchez, 2020

Need-eligible students subsample

Londoño-Vélez et al. (2020)

Figure: Immediate Postsecondary Enrollment



Source: Londoño-Vélez, Rodríguez, and Sánchez, 2020

Merit-eligible students subsample

Londoño-Vélez et al. (2020)- Model specification

- Y_i: post-secondary enrollment
- X_i: score in SABER 11 (or SISBEN)

- The average outcome given the score is:

$$\mathbb{E}\left[Y_i \mid X_i\right] = \begin{cases} \mathbb{E}\left[Y_i(0) \mid X_i\right] & \text{if } X_i < 0 \\ \mathbb{E}\left[Y_i(1) \mid X_i\right] & \text{if } X_i \geq 0 \end{cases}$$

- In the RD design, the treatment effect is defined as:

$$\tau_{SRD} \equiv \mathbb{E}\left[Y_i(1) - Y_i(0) \mid X_i = 0\right]$$

- Identification assumption: $\mathbb{E}\left[Y_i(1) \mid X_i = 0\right]$ and $\mathbb{E}\left[Y_i(0) \mid X_i = 0\right]$ are continuos at $x = 0$.

$$\tau_{\text{SRD}} = \lim_{x \downarrow 0} \mathbb{E}\left[Y_i \mid X_i = x\right] - \lim_{x \uparrow 0} \mathbb{E}\left[Y_i \mid X_i = x\right]$$

Londoño-Vélez et al. (2020)

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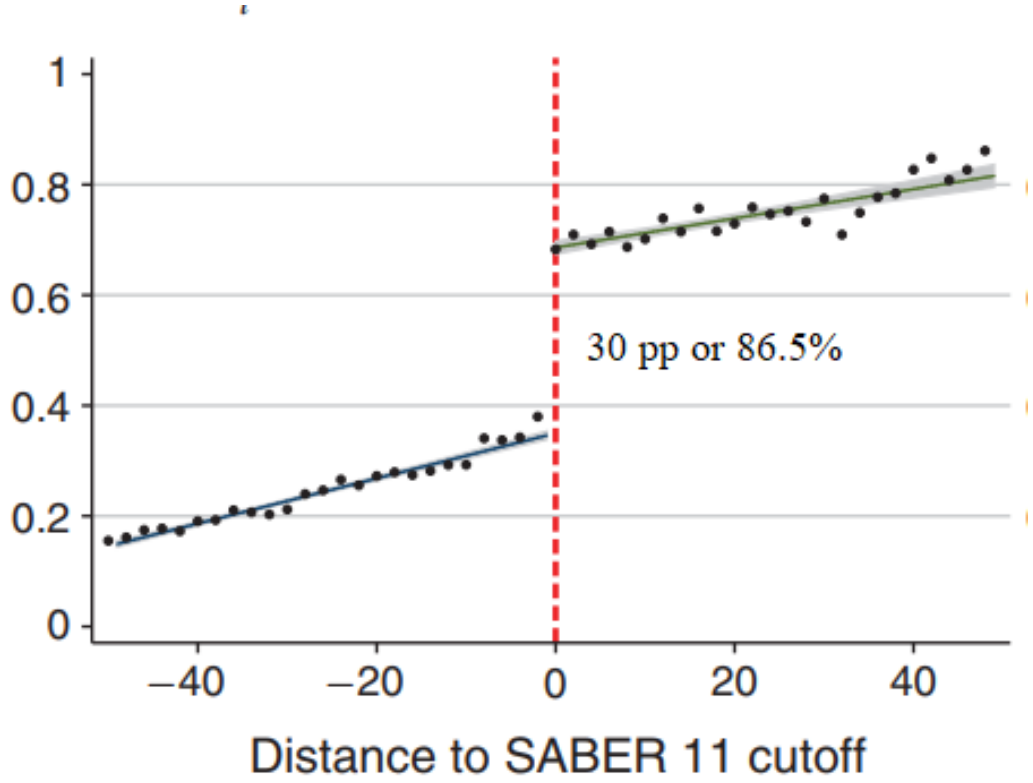

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Need-eligible students subsample

Londoño-Vélez et al. (2020)

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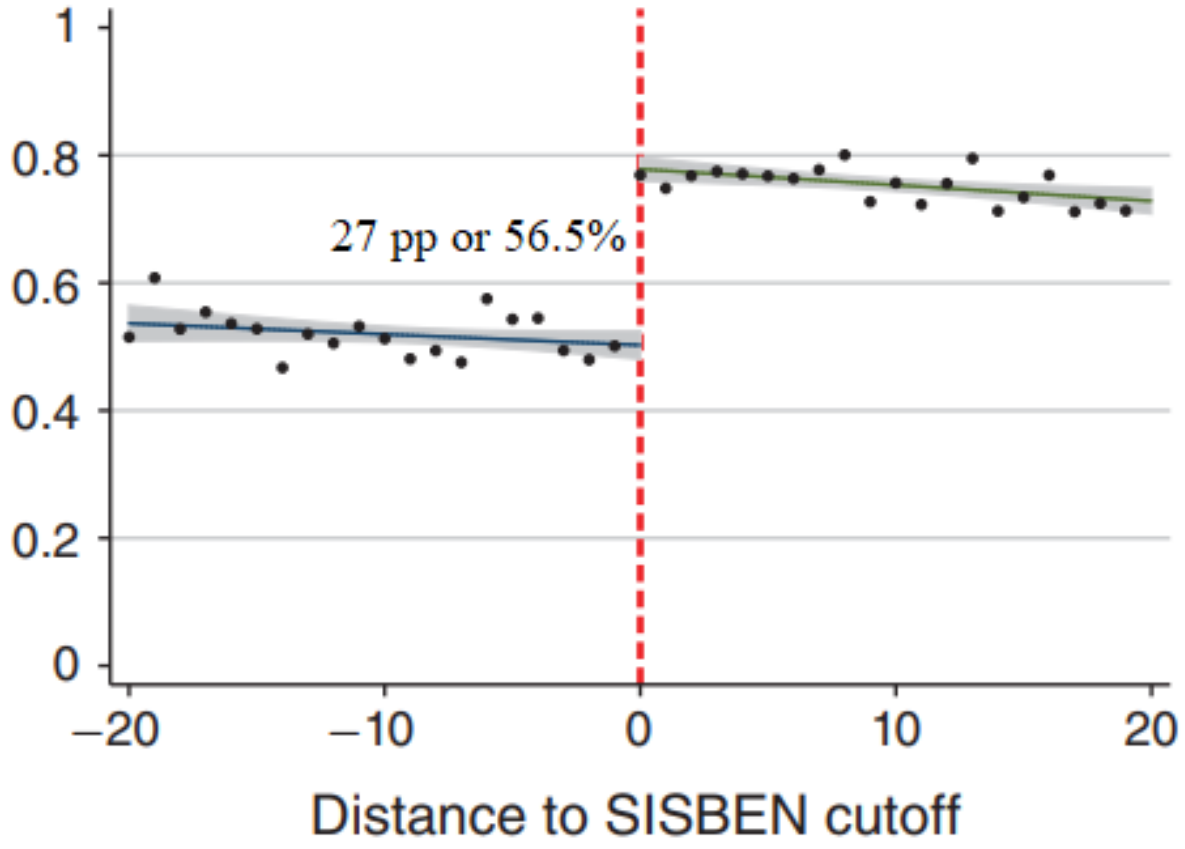


Source: Londoño-Vélez, Rodríguez, and Sánchez, 2020

Merit-eligible students subsample

My project

1. Heterogeneity analysis
2. Robustness check

Heterogeneity analysis

Goal: identify the presence of heterogeneity in the effect of removing the financial constraints on low-income, high-achieving students by gender, age, ethnicity and employment status.

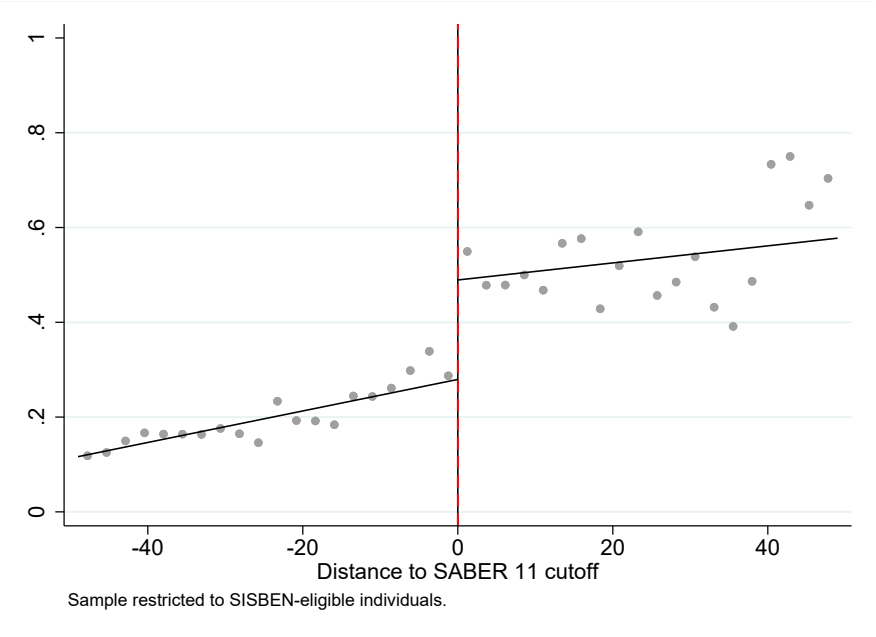
- I perform the same estimations that the authors in 8 subsamples following 4 dummy variables
- Let $J = \{Female, Age_over_average, Employed, Ethnic_minority\}$
- For each $j \in J$ I define:

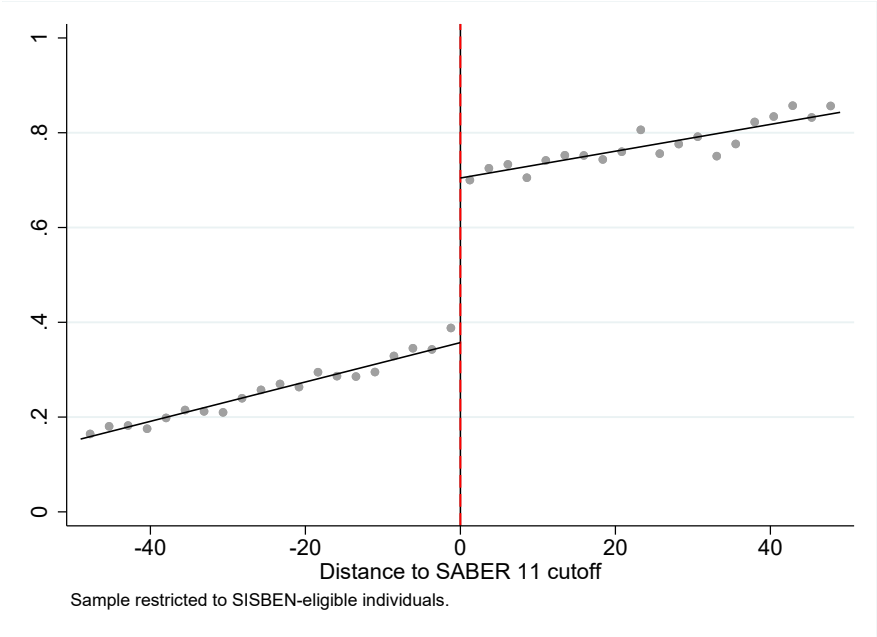
$$\tau_j^{DIFF} = \tau_{j=1} - \tau_{j=0}$$

- Standard errors are calculated using non-parametric bootstrap

Heterogeneity analysis-Results

Figure: Immediate Post-secondary Enrollment by Employment Status





Heterogeneity analysis-Results

- Results:

	Gender	Age	Employment	Minority
Panel A. SABER 11 as the running variable				
τ^{DIFF}	0.003	-0.086	-0.153	-0.073
	(0.031)	(0.049)	(0.051)	(0.058)
Panel B. SISBEN as the running variable				
τ^{DIFF}	0.085	0.068	-0.044	-0.003
	(0.071)	(0.115)	(0.133)	(0.129)

*

My project

1. Heterogeneity analysis
2. Robustness check

Robustness check

Goal: identify the average treatment effect on all treated students, by building a control group with the students who took the SABER 11 exam in 2013 and would be eligible to receive treatment.

- Model: *

$$Y_i = \beta_0 + \beta_1 D_i + X_i' \delta + \varepsilon_i$$

$D_i = 1$ if the student belong to the treated group (cohort 2014)

- Crucial assumption: the only difference between both cohorts is the students' year of birth

Robustness Check results

	Model (1)	Model (2)
Effect of being eligible for financial aid	0.281	0.300
	(0.004)	(0.022)
Control variables		X
Observations	28,259	25,778

Final remarks

- Only the heterogeneity by employment status for need-eligible students is statistically significant. Why?
- The robustness check shows that the treatment effect estimated using as control group eligible students who took the SABER 11 exam in 2013 are in line with the author's estimations.

Thank you!

References

Londoño-Vélez, Juliana, Catherine Rodríguez, and Fabio Sánchez (2020). "Upstream and Downstream Impacts of College Merit-Based Financial Aid for Low-Income Students: Ser Pilo Paga in Colombia". In: *American Economic Journal: Economic Policy* 12.2, pp. 193–227. DOI: 10.1257/pol.20180131. URL: https://www.aeaweb.org/articles?id=10.1257/pol.20180131.

RD estimates

Table: Immediate Enrollment in Postsecondary Education - sub-samples.

Sample	All	Female	Male	Old	Young	Work	No Work	Minor.	No Minor.
Panel A. SABER 11 as the running variable									
RF	0.32	0.321	0.318	0.244	0.330	0.175	0.328	0.252	0.325
	(0.012)	(0.018)	(0.015)	(0.042)	(0.012)	(0.044)	(0.013)	(0.045)	(0.013)
Mean control	0.37	0.381	0.361	0.293	0.380	0.337	0.375	0.435	0.364
Observations	299475	172182	127293	85420	214055	40830	258645	41525	257950
BW loc. Poly.	29.679	27.434	33.501	22.072	34.205	26.827	27.422	28.243	29.523
Effect obs. Control	31170	14076	18874	2572	34537	2520	25494	2812	28244
Effect obs. Treat	11711	5098	6905	1105	11225	1010	10330	898	10803
Panel B. SISBEN as the running variable									
RF	0.274	0.323	0.238	0.337	0.269	0.232	0.276	0.274	0.277
	(0.027)	(0.042)	(0.032)	(0.082)	(0.029)	(0.099)	(0.029)	(0.117)	(0.029)
Mean control	0.485	0.438	0.528	0.308	0.500	0.368	0.494	0.592	0.476
Observations	23132	9980	13152	2129	21003	1961	21171	1557	21575
BW loc. Poly.	9.028	8.502	11.996	12.660	8.756	9.370	8.871	8.650	8.487
Effect obs. Control	3868	1577	2844	352	3525	273	3562	200	3476
Effect obs. Treat	3902	1582	2914	446	3476	323	3521	249	3391

Stats Robustness Check

	Eligible students			
	2013		2014	
	mean	sd	mean	sd
Immediate enrollment -HEI	0.460	0.498	0.741	0.438
Female	0.417	0.493	0.429	0.495
Age	16.300	1.298	16.610	2.313
Age-Over average	0.078	0.268	0.076	0.265
Ethnic minority	0.317	0.465	0.299	0.458
Household SES: stratum 1	0.447	0.497	0.451	0.498
Household SES: stratum 2	0.216	0.411	0.221	0.415
Household SES: stratum 3	0.017	0.129	0.023	0.151
Household SES: stratum 4	0.003	0.054	0.004	0.066
Household SES: stratum 5	0.000	0.022	0.001	0.028
Household SES: stratum 6	0.116	0.320	0.214	0.410
Mother education: primary	0.452	0.498	0.459	0.498
Mother education: secondary	0.262	0.440	0.157	0.363
Mother education: T&T	0.170	0.376	0.171	0.376
Mother education: higher	0.176	0.381	0.289	0.453
Father education: primary	0.412	0.492	0.416	0.493
Father education: secondary	0.235	0.424	0.119	0.324
Father education: T&T	0.177	0.381	0.176	0.381
Father education: higher	4.611	1.612	4.528	1.543
Family size	0.980	0.139	0.092	0.288
Employed	0.202	0.402	0.216	0.412
Private school	0.604	0.489	0.594	0.491
School hours: morning	0.006	0.076	0.008	0.088
School hours: evening	0.162	0.369	0.163	0.370
School hours: afternoon	0.004	0.064	0.007	0.081
School hours: weekends	0.061	0.238	0.102	0.303
Observations	12,883		15,423	